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Paleomagnetism of Mesoproterozoic Satakunta sandstone, Western Finland

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ABSTRACT

A paleomagnetic study of the Mesoproterozoic Satakunta sandstone formation, Western Finland is presented here. Two components of natural remanent magnetization (NRM) were isolated in the sandstone with alternating field (AF) and thermal demagnetization treatments. The first is a high coercivity/unblocking temperature component which, after tilt and inclination shallowing corrections, yields a remanent magnetization of $D = 25.2^\circ$, $I = 3.9^\circ$ with $\alpha_{95} = 9.1^\circ$ (10 sites), corresponding to a paleomagnetic pole of Plat = 27.8° N, Plon = 173.2° E, with $A_{95} = 6.5^\circ$. Positive reversal and tilt tests, an inverse baked contact test, as well as a stratigraphic order of the site mean poles, indicates a primary remanence. The relative age of the sandstone is estimated at ca. 1600 based on the proximity to other well defined poles of 1540–1770 Ma from Baltica. However, because the pole also overlaps with poles with the ages of 1452–1630 Ma, the age of the sandstone cannot be accurately defined. The second component is a lower coercivity/unblocking temperature component with a remanent magnetization direction $D = 34.0^\circ$, $I = -40.7^\circ$ with $\alpha_{95} = 8.1^\circ$ (13 sites) similar to that of the Postjotnian Satakunta diabase intrusions, which are widespread in the area. Comparing APWPs of Baltica and Laurentia during 1.77–1.27 Ga gives strong support to the Mesoproterozoic NENA configuration at 1.6 Ga.

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1. Introduction

The paleographic position of Baltica and its relationship to other continental blocks during the Proterozoic are based on little data, and are often of limited reliability (Pesonen et al., 2003, 2012). Most cratonic blocks show evidence of either collision or accretion between 1.9 and 1.8 Ga, which lead researchers to propose the Mesoproterozoic supercontinent Columbia (Hudsonland, Nuna) (Meert, 2002; Pesonen et al., 2003; Zhao et al., 2002, 2004). According to several authors (e.g. Gower et al., 1990; Salminen and Pesonen, 2007; Evans and Pisarevsky, 2008; Lubnina et al., 2010; Evans and Mitchell, 2011; Salminen et al., 2014) Baltica and Laurentia were connected in the NENA (North Europe-North America) configuration forming the core of the Columbia supercontinent between 1.8 and 1.2 Ga. However there are not only significant gaps in the apparent polar wander paths (APWP's) of Baltica and Laurentia between 1.78 Ga and 1.27 Ga, but also controversies in attempts to match the APWP's (Salminen et al., 2014).

Although there are numerous paleomagnetic poles from the Mesoproterozoic for Baltica, many of them fulfil very few of the reliability criteria set out by Van der Voo (1990). A case in point is the earlier study on the Mesoproterozoic Satakunta sandstone (Neuvonen, 1973). Old-fashioned demagnetization methods ("blanket cleaning") were used, which are insufficient to isolate the characteristic natural remanence (ChRM). Furthermore, an insufficient number of samples ($N = 10$) were used for the study, no reversals were observed from the four study sites, and no tilt- or inclination-corrections were applied. Moreover, the results are doubted since it closely resembles the results of Postjotnian diabases (Neuvonen, 1965).

In this study we revisited the Mesoproterozoic Satakunta sandstone, systematically sampling all the known sandstone outcrops, and using modern demagnetization techniques coupled with rock magnetic measurements. The objective is to identify the ChRM of the Satakunta sandstone, to estimate the relative paleomagnetic age of deposition, and to use the new paleomagnetic pole for Baltica to check the validity of NENA at the estimated age of the sandstone.

2. Geological background

Satakunta sandstone is a fluvial formation situated in the Satakunta region in Western Finland (Fig. 1). This unmetamorphosed sandstone was deposited in a deltaic environment and has been preserved in a SE-NW trending graben within the

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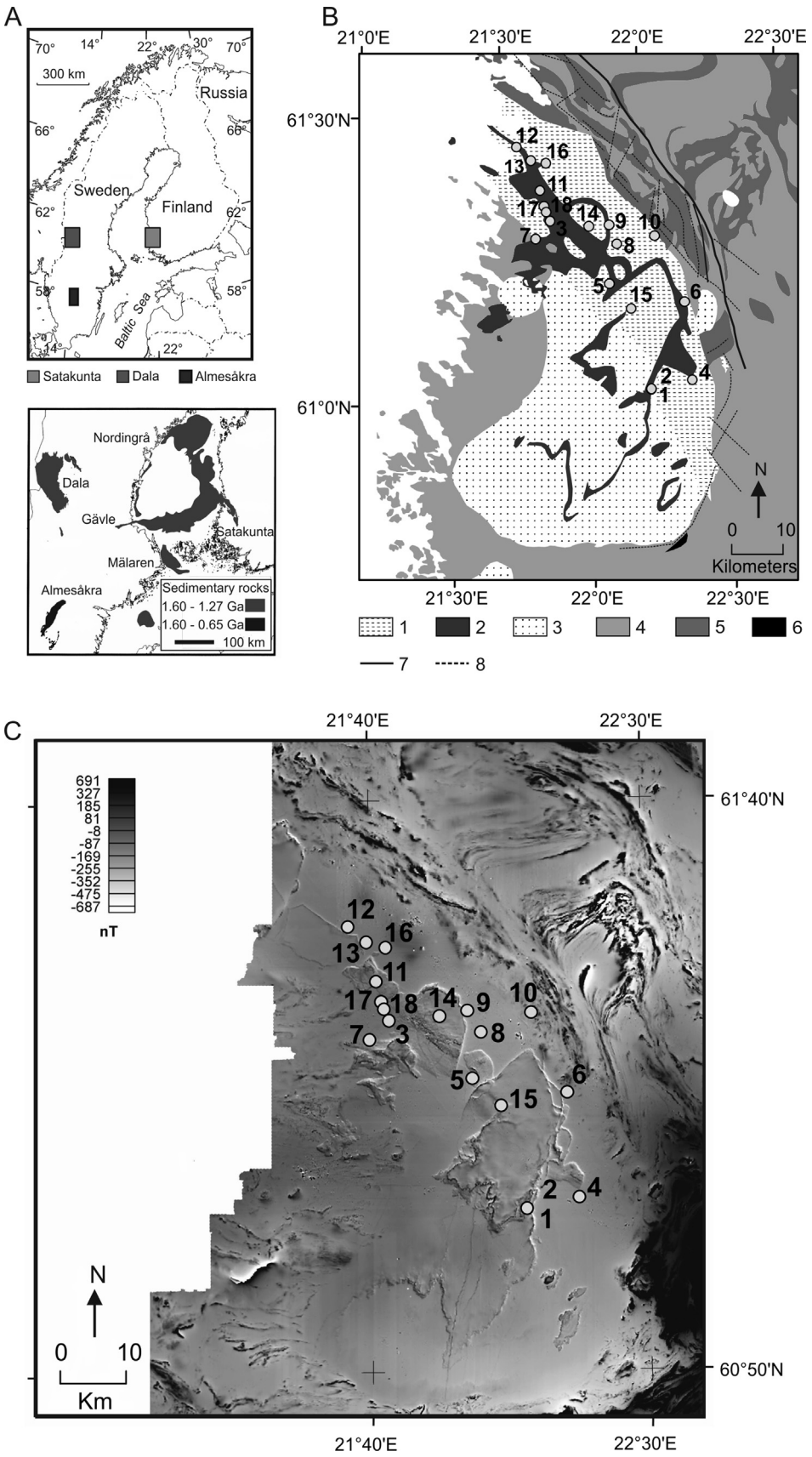


Fig. 1. (A) Satakunta sandstone and other Fennoscandian sedimentary formation. (B) Bedrock of Satakunta: 1. Sandstone, 2. Postjotnian Diabase, 3. Rapakivi granite, 4. Svecofennian bedrock (Southern Finland arc complex), 5. Svecofennian bedrock (Central Finland arc complex), 6. Gabbro, 7. Kynsikangas shear/fault zone, 8. Faults/shears (modified from Pajunen and Wennerström (2010)). Studied sandstone sites (and sample codes used in this study). 1. Kiperi (80-KP, KT), 2. Kiperoja (80-KO, KJ), 3. Sahankoski (SK), 4. Karhusuo (MU, SS, SY), 5. Panelia (PA), 6. Koomankallio (KK), 7. Naskalinkallio (NK), 8. Leistilänjärvi (LJ), 9. Leistilä (LE, LS), 10. Harjavalta (HA, HC, HV), 11. Knapernummi (KM, KN), 12. Makholm (MH), 13. Kalliopää (KP), 14. Pössi (PS), 15. Kiukainen (KI), 16. Metsäkulma (MK), 17. Murro-oja (MO), and 18. Kotkajärvenoja (KR). All sites marked on the map are sandstone outcrop sites; diabase was collected from outcrops in contact with or in close proximity to the sandstone outcrops. (C) Aeromagnetic anomaly map of Satakunta (copyright: GTK airborne geophysical database). Nominal terrain clearance is 30 m and line spacing is 200 m. Flight line direction mainly E–W, partly N–S.

Paleoproterozoic Svecofennian Domain. It is part of a much wider sandstone basin extending across the Baltic Sea to the Gävle valley and Nordingrå area in Sweden (Fig. 1; Kohonen et al., 1993; Pokki et al., in press). The Laitila rapakivi batholith (1570 Ma; Vaasjoki, 1977) is situated to the south of the sandstone. Three minor rapakivi stocks – Eurajoki, Kokemäki and Reposaari – are situated to the southwest, northeast and northwest, respectively. The contact between the sandstone and rapakivi is a result of faulting, as the emplacement of the rapakivi took place ca. 10–30 km below the basin (Pokki et al., in press). This is supported by grain size measurements from sandstone outcrops and drill cores showing coarse grain, unmetamorphosed sediments close to the southern margin (Kohonen et al., 1993). The sandstone and rapakivi granite are cut, vertically and horizontally, by Postjotnian olivine diabase dykes and sills of 1264 ± 12 Ma in age (Suominen, 1991). The sandstone is poorly exposed and mostly overlain by Quaternary deposits.

At the present erosion level the Satakunta formation consists of red-coloured subarkosic (at the southwestern margin) and arkosic (in other parts) sandstone (Pokki et al., in press, and references therein) with thin intercalated beds of red and black shales (Simonen and Kouvo, 1955). The sandstone contains generally between 40–80% quartz and 15–30% feldspar (Simonen and Kouvo, 1955) and originated from the surrounding Svecofennian supracrustal and plutonic rocks. The magnetization of the sandstone is carried by magnetite and hematite (Neuvonen, 1973). The hematite is both detrital as well as a product of biotite alteration (Marttila, 1969). The sandstone formation is ca. 1300 m thick in the Pori region in the West, decreasing towards the southeast (Elo and Pirttijärvi, 2010).

Block movements after the sandstone formation caused tilting of the sandstone beds, particularly in the northeastern part of the graben (Kohonen et al., 1993). The tilting, with a dip of ca. 5° (ranging between 0° and 40°) to North and Northeast (Pokki, 2007), implies that the outcrops in the South and Southwest are older (lower in the stratigraphy) than in the Northeast.

The exact age of the sandstone deposition is unknown, however it is certain that deposition took place between the Subjotnian rifting event at ca. 1500–1650 Ma (Kohonen and Rämö, 2005) (when several rapakivi intrustions and related dyke swarms were emplaced across the southern Fennoscandian Shield), and the Postjotnian intrusion event at 1.26 Ga. A ^{40}K–^{40}Ar age for intercalated red shale at two outcrops yielded an age of 1300 Ma and 1150 Ma, respectively (Simonen, 1960). However, this represents the age of diagenesis of the shale, and is probably, as Simonen (1960) proposed, affected by the Postjotnian diabase intrusions. Neuvonen (1973) reported a paleomagnetic APWP age of 1400–1300 Ma, however, since the present study is a re-evaluation of the paleomagnetism of Satakunta sandstone, this paleomagnetic age is also under scrutiny.

3. Methods

3.1. Sampling and sample preparation

Sandstone and Postjotnian diabase samples were collected from eighteen sites within the Satakunta sandstone formation (Fig. 1). At the sandstone sites where diabase is exposed, the diabase was also sampled. Some of the samples were taken with a Pomeroy portable field drill, but the majority of the samples were taken as block samples. The samples were oriented with a magnetic compass and the local declination was corrected. Multiple specimens were prepared from each sample; block samples were drilled in the laboratory and all the drill cores were sawed into cylinders (2.5 cm diameter (D), 2.2 cm length (h), with $h/D = 0.9$).

3.2. Laboratory measurements

The measurements were carried out at the Solid Earth Geophysics Laboratory of the University of Helsinki. Magnetic susceptibility, intensity of natural remanent magnetization (NRM), and density were measured for each specimen using the Risto-5 petrophysics assembly. Paleomagnetic measurements with progressive alternating field and thermal demagnetizations were performed using a 2G-DC SQUID magnetometer and MMTD1 thermal demagnetizer. During thermal cleaning, magnetic susceptibility was measured after each heating step, to monitor chemical alterations. To minimize oxidation during thermal demagnetization, some samples were heated in an Argon atmosphere. Several samples underwent low temperature magnetic cleaning by immersion in liquid nitrogen prior to thermal treatments (Borradaile et al., 2004).

NRM components were visually identified using stereographic and orthogonal projections (Zijderveld, 1967) and the directions were calculated by a least squares method (Leino, 1991), with a mean angular deviation (MAD) of $< 6°$ (with a few coarser grained samples a larger MAD was used). Mean remanence directions for the different components were calculated according to Fisher (1953). Tilt correction calculations were done using the Super-IAPD programme and paleoreconstructions were plotted with the GMAP programme (Torsvik and Smethurst, 1999). To determine if the paleosecular variation is averaged out, the angular dispersion for the pole (S) was calculated for each sandstone site carrying the sandstone component as well as for the mean diabase component using the equation:

$$S = \frac{81^\circ}{K^{1/2}}$$

where K is the Fisher distribution of the poles.

Thermomagnetic (susceptibility vs. T) and magnetic hysteresis measurements were carried out on one sample from each site to identify the magnetic carriers. The former was measured for mineral identification with the AGICO-KLY-3S kappabridge, which operates at a frequency of 875 Hz and a field intensity of 300 A m^{-1}. Each sample was heated in argon gas from room temperature up to 700 °C and cooled back to room temperature. Magnetic hysteresis properties were measured using a MicroMag^{TM}3900 Vibrating Sample Magnetometer (VSM) by Princeton Measurement Corporation with a maximum field of 1 T.

High field anisotropy of isothermal remanence magnetization (hf-AIR) measurements were carried out to correct inclination shallowing of the sandstone. The hf-AIR measurement technique of Bilardello and Kodama (2009) was followed; 5 T IRM's were applied in 9 orientations (Girdler, 1961) of 9 mm × 7 mm oriented samples, using a MMPM 10 pulse magnetizer. Samples were then thermally demagnetized at 125 °C in a MMTD1 thermal demagnetizer, to demagnetize goethite. The samples were then AF demagnetized at 100 mT, to demagnetize magnetite. The remaining hematite component was measured in a SQUID magnetometer. The best-fit anisotropy tensor was determined using the least square fitting, and the eigenvalues were calculated from the tensor. The inclination correction was applied by using the hematite correction equation of Tan and Kodama (2003):

$$\frac{\tan I_c = \tan I_m (2a+1)k_{max} - 1}{(2a+1)k_{min} - 1}$$

where I_c and I_m are the corrected and measured inclinations, respectively, k_{max} and k_{min} are the normalized maximum and minimum principal axes, respectively of the fabric, and a is the individual particle anisotropy. The latter was not measured, but a value of 1.4 was used, which is considered to be reasonable for hematite (Tan et al., 2007; Kodama, 2012)

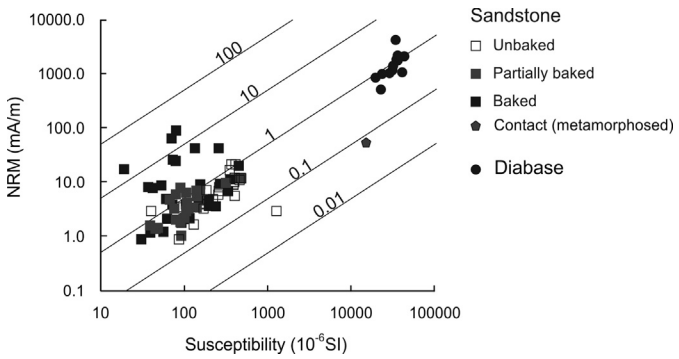


Fig. 2. Magnetic susceptibility vs. NRM (logarithmic scales) for Satakunta sandstone and diabase samples. The Koenigsberger (*Q*) ratio is shown as diagonal lines.

4. Results

4.1. Petrophysical results

Petrophysical properties (density, susceptibility, NRM and *Q*-value) for the sandstone and the diabase are summarized in Table 1. An NRM vs. susceptibility plot is shown in Fig. 2. As can be expected, there is a very clear distinction between diabase and sandstone. The sandstone samples are further grouped as unbaked and baked or partially baked as defined from a baked remanent magnetization component originating from the heat caused by nearby ca. 1.26 Ga diabase intrusions. Unbaked sandstone samples do not contain a baked component, partially baked samples carry both a baked component as well as the proposed primary sandstone component, and baked samples are completely overprinted with the baked component. The baked sandstone samples show higher variability in NRM with some samples showing significantly higher NRM values, consistent with thermo-chemical remanent magnetization acquisition during baking. One sample from Leistilänjärvi, marked as 'metamorphosed' in Table 1 and Fig. 2, was collected from the very contact between sandstone and diabase. The petrophysics of this sample show values distinct from the other sandstone samples possibly because of physiochemical alteration of minerals due to baking.

4.2. Rock magnetic results

Fig. 3 shows thermomagnetic curves (susceptibility vs. temperature) of unbaked, partially baked and baked sandstone as well as diabase. The heating curves for unbaked sandstone samples show the presence of goethite, magnetite and hematite, and the cooling curve indicates the formation of magnetite during heating. Baked sandstone samples show goethite and magnetite less distinctly (or not at all) and show a partially reversible cooling curve, indicating that magnetite was formed in situ, possibly caused by the heat from the diabase intrusions. Partially baked samples show a similar trend as baked samples, with a less distinct Hopkinson peak in the heating curve. The thermomagnetic curves for diabase clearly show the presence of magnetite in the heating curve.

Fig. 4 shows the hysteresis loops of the studied samples. Most of the sandstone samples were not saturated at the maximum field of 1 T and showed a wasp waisted curve, indicating the presence of higher coercivity magnetic minerals such as hematite and/or goethite and a lower coercivity magnetite. The Postjotnian diabase sample saturated below 0.3 T, indicating magnetite, which is the main magnetic mineral in the diabase (Neuvonen, 1965). The rock magnetic results of the sandstone are in agreement with mineralogy studies summarized in Pokki et al. (in press).

4.3. Paleomagnetic results

Paleomagnetic results from Satakunta sandstone and diabase sheets are shown in Table 2 and examples of AF and thermal demagnetization behaviour are shown in Figs. 5 and 6. The mean paleomagnetic directions are plotted in Fig. 8.

The sandstone carries two prevailing components. The first component (SS) was obtained from 10 sites and is of dual polarity (component SS_N and SS_R, the former obtained from 9 sites, and the latter obtained from 3 sites; Table 2) and is directed to the NNE (SSW), and has a shallow inclination. The other component in sandstone (SS_B) is of normal polarity and is directed to the NE and with a much steeper, negative inclination. This component is similar to the component carried by the Postjotnian Satakunta diabase dykes (DB) as well as other Postjotnian directions seen in Finland and Sweden (Neuvonen, 1965; Elming and Mattsson, 2001). Unbaked sandstone specimens (e.g. PA3-3B in Fig. 5) carry the SS component together with a viscous component, which is usually a present Earth field (PEF) component. Fully baked sandstone specimens (e.g. KJ3-2A in Fig. 5) contain only the SS_B component. Partially baked sandstone specimens (e.g. SK6-2A in Fig. 5) carry both SS and SS_B components.

Thermal demagnetization of sandstone reveals a PEF component at low unblocking temperatures and the sandstone component (SS) at intermediate to high unblocking temperatures. The SS_N component in specimen SY4-1A (Fig. 6) is isolated between unblocking temperatures of 200–680 °C. In some cases the diabase component (SS_B) is revealed at low to intermediate unblocking temperatures. Thermal demagnetization of diabase (e.g. KT2-1C in Fig. 6) shows that the DB component is carried by magnetite.

The general trend in demagnetization of partially baked specimens is a movement from the steeper diabase direction (lower coercivities/unblocking temperatures) to the shallower sandstone component (higher coercivities/unblocking temperatures).

4.4. Bedding tilt and inclination corrections

The Satakunta sandstone formation has undergone structural tilting with variable dip angles observed at the different sandstone outcrop sites (Table 3). The dip direction varies, but has a general northern to northeastern trend.

Table 1
Petrophysical properties of sandstone and diabase.

Site	$B^*/N/n$	D (kg/m^3)	k (10^{-6} SI)	J_r (mA m^{-1})	Q
Sandstone					
Unbaked	7*/25/58	2582 ± 25	308 ± 219	6.4 ± 3.8	0.8 ± 0.2
Partially baked	7*/25/50	2600 ± 45	186 ± 151	5.6 ± 3.6	0.9 ± 0.3
Baked	11*/31/57	2604 ± 23	139 ± 121	23.2 ± 39.0	4.7 ± 5.6
Contact (metamorphosed)	1/1	2728	15,020	54.2	0.1
Diabase	8*/13/28	2921 ± 99	33,108 ± 4471	1797.1 ± 1045.9	1.3 ± 0.7

$B/N/n$ is the number of sites/samples/specimens; D is density; k is magnetic susceptibility; J_r is the intensity of NRM; Q is the Koenigberger ratio; the standard deviations are for the mean value for each site.

* Statistical level used for mean calculations.

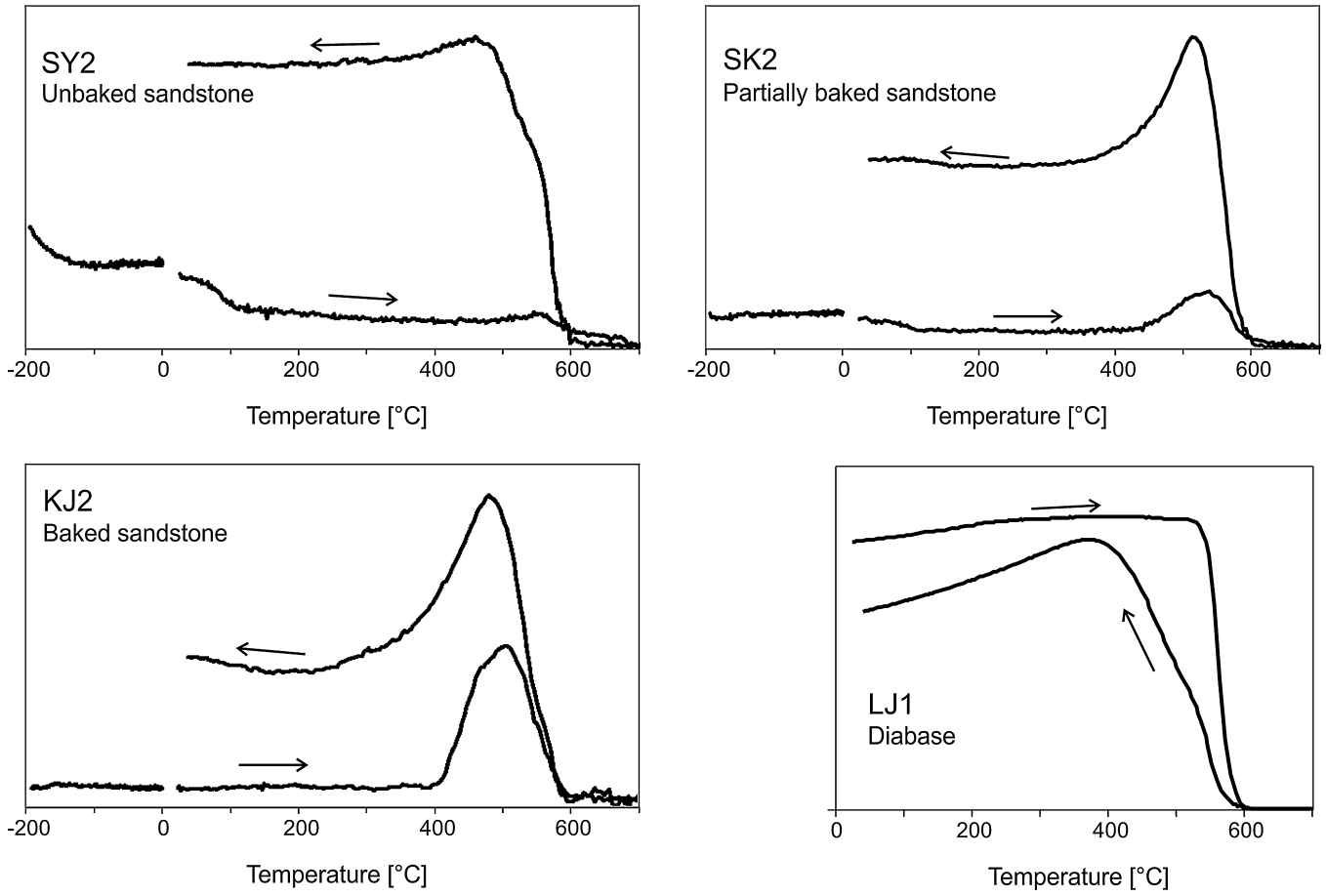


Fig. 3. Thermomagnetic curves (susceptibility vs. temperature) for Satakunta sandstone and diabase samples.

The tilt correction applied to the SS component for each site results in a decrease in α_{95} and an increase in the k value for the mean direction, indicating a pre-tilt acquisition of the paleomagnetic direction. The Harjavalta site (HC, HA and HV in Fig. 1) has a pre-tilting direction similar, and of opposite polarity, to the baked direction due to steep tilting, and therefore has a large influence on the tilt correction test. The tilt correction test was repeated without this site direction, and still gave improved α_{95} and k values, but less pronounced however. Furthermore, the Harjavalta site has a reversed direction, and as there are no reversed Postjotnian directions in Satakunta, the pre-tilt direction at Harjavalta is unlikely to be a baked direction. Since the site directions have both negative and positive shallow inclinations (Fig. 7), it must be noted that the inclination correction causes the site means to become more scattered (from shallow upwards and downwards directions to steeper upward and downward directions), and therefore the mean direction did not change significantly, and the α_{95} and k values did not improve.

The hf-AIR measurements of the sandstone samples yielded an average flattening factor of 0.64. The paleomagnetic direction of the SS component, after tilt- and inclination- corrections, is $D = 25.2^\circ$; $I = 3.9^\circ$; $\alpha_{95} = 9.1^\circ$; $k = 28.9$ and is shown in Table 3 and Fig. 7. The corrected site directions pass the McFadden and McElhinny (1990) reversal test with classification Rc ($\gamma = 16.3^\circ$ and $\gamma_c = 24.6^\circ$).

5. Discussion

We interpret the SS component as the primary sandstone component due to the positive reversal and tilt tests. Component SS_B is likely a baked component formed by heating from Postjotnian diabase intrusions. This is supported by the reversal in thermomagnetic curves (Fig. 3), and increase in NRM (Fig. 1) of baked and partially baked samples, indicating in situ heating, and subsequent oxidation of these rocks. Furthermore, the SS_B component shows a similar paleomagnetic direction to other Postjotnian formations (Neuvonen, 1965; Elming and Mattsson, 2001). Baked sandstone specimens (Table 2), containing both SS_B and SS components (for examples, see Figs. 5 and 6, specimens SK6-2A and KJ5-3C, respectively), fulfil a baked contact test for the diabase, and therefore an inverse baked contact test for sandstone.

Paleomagnetic directions from the earlier study on Satakunta sandstone by Neuvonen (1973), were similar to the Postjotnian diabase direction. Single step demagnetization of 400 °C was applied to sandstone samples, which is lower than the unblocking temperatures of the magnetite and hematite carrying the sandstone direction. This lead us to believe that the sandstone direction obtained by Neuvonen ($D = 24^\circ$; $I = -41^\circ$) is in fact a baked component (SS_B) resulting from baking by adjacent diabase intrusions.

5.1. Paleomagnetic poles

The primary (combined N and R) sandstone component, SS ($D = 25.2^\circ$; $I = 3.9^\circ$; $\alpha_{95} = 9.1^\circ$; $k = 28.9$; 10 sites; Table 3) yields a paleomagnetic pole of Plat = 27.8° N, Plon = 173.2° E with $A_{95} = 6.5^\circ$. The baked sandstone component, SS_B ($D = 34.0^\circ$; $I = -40.7^\circ$; $\alpha_{95} = 8.1^\circ$; $k = 27.5$; 13 sites; Table 2), yields a paleomagnetic pole of Plat = 1.2° N, Plon = 171.1° E, $A_{95} = 7.6^\circ$. From the diabase sites, we obtained a mean direction of $D = 44.0^\circ$; $I = -52.3^\circ$; $\alpha_{95} = 8.3^\circ$; $k = 45.8$; 8 sites (Table 2), and a paleomagnetic pole of Plat = 11° S, Plon = 166° E, $A_{95} = 9^\circ$. Although the baked sandstone component

Table 2
Mean paleomagnetic data for sandstone and diabase.

Sites	Glat (°N)	Glon (°E)	p	(*B*)/*N*/*n*	*D* (°)	*I* (°)	α_{95} (°)	*k*	*S*
Sandstone, component SS									
MU, SS, SY	61.09	22.29	N	11*/28	23.6	7.3	10.8	18.8	18.0
SK	61.35	21.73	N	3*/8	25.6	5.8	12.4	99.9	5.3
PA	61.25	21.96	N	5*/18	26.1	3.9	7.8	97.4	5.3
LE, LS	61.35	21.95	N	1/2	33.4	8.0			
KJ	61.07	22.15	N	3*/6	20.5	18.6	17.9	48.6	12.1
KK	61.23	22.24	N	1/1	22.5	12.7			
NK	61.32	21.69	N	3*/4	28.3	15.3	15.7	63.0	9.8
KT	61.08	22.15	N	1/3	356.1	27.6			
LJ	61.32	21.98	N	2/2	36.6	−13.8			
LJ	61.32	21.98	R	1/1	204.4	−20.2			
HA, HC, HV	61.34	22.12	R	3*/5	216.8	33.4	24.2	27.0	14.3
LE	61.35	21.95	R	3*/10	215.6	7.4	13.3	87.3	7.1
Mean sandstone (normal)			N	(9*)/30/72	24.0	9.7	10.2	26.5	12.9
Mean sandstone (reversed)			R	(3*)/7/16	212.0	6.9	44.5	8.7	16.0
Grand mean			C	(10*)/37/88	24.9	5.5	12.0	17.2	14.4
Baked sandstone, component SS_B									
KJ	61.07	22.15	N	8*/25	26.2	−38.5	4.8	136.3	
KK	61.23	22.24	N	1/2	61.7	−31.4			
KM, KN	61.40	21.69	N	3*/3	31.1	−41.7	9.4	173.9	
LS	61.35	21.95	N	2/2	21.0	−39.7			
MU, SS, SY	61.09	22.29	N	8*/19	32.1	−29.0	10.1	31.2	
SK	61.35	21.73	N	4*/9	24.7	−37.6	15.9	34.4	
PS	61.35	21.87	N	4*/5	19.5	−46.9	4.9	349.7	
PA	61.25	21.96	N	1/2	22.5	−32.2			
NK	61.32	21.69	N	7*/14	17.7	−43.6	8.2	54.6	
LJ	61.32	21.98	N	2/2	61.6	−44.7			
KT	61.08	22.15	N	6*/23	14.4	−42.7	7.1	91.1	
KI	61.21	22.05	N	1/3	47.2	−31.5			
KP	61.61	21.63	N	3*/3	68.2	−52.3	8.3	220.0	
Mean baked sandstone				(13*)/50/112	34.0	−40.7	8.1	27.5	17.6
Diabase, component DB									
80-KO	61.07	22.15	N	2/2	50.9	−50.3			
80-KP	61.08	22.15	N	2/2	32.3	−61.5			
KT	61.08	22.15	N	1/6	50.1	−53.6			
NK	61.32	21.69	N	1/7	38.9	−59.2			
MH	61.47	21.59	N	1/1	44.2	−39.1			
MM	61.39	21.86	N	3*/3	31.4	−40.7	21.9	32.8	
99-KO	61.37	21.80	N	2/6	61.9	−44.4			
LJ	61.32	21.98	N	1/1	36.5	−66.1			
Mean diabase				(8*)/13/28	44.0	−52.3	8.3	45.8	13.0
Mean (baked sandstone + diabase)				(21*)/63/140	37.4	−45.2	6.2	27.7	17.0

Glat, Glon = geographical latitude and longitude (a reference location of 61.2° N, 22° E was used for grand mean directions); p = polarity where N = normal, R = reversed, C = combined; *B*/*N*/*n* = number of sites/samples/specimens where * indicates statistical level used for mean calculations; *D*, *I* = declination, inclination of the remanent magnetization component; α_{95} (°) = the radius of the 95% confidence circle of the mean direction; *k* = precision parameter (Fisher, 1953); *S* = angular dispersion of paleomagnetic poles.

Table 3
Paleomagnetic results of sandstone before tilt correction, after tilt correction and after inclination shallowing correction.

Site	Pol.	Az.	Before tilt correction					After tilt correction				After inclination correction						
			Dip	*D*	*I*	α_{95}	*k*	*D*	*I*	α_{95}	*k*	*D*	*I*	α_{95}	*k*	Plat	Plon	A_{95}
MU, SS, SY	N	300	5	23.6	7.3			23.6	2.3			23.6	3.8			28	175	8
SK	N	340	10	25.6	5.8			25.9	−1.4			25.9	LS.2.1			25	173	9
PA	N	0	2	26.1	3.9			26.2	3			26.2	5.3			28	172	6
LE, LS	N	115	11	33.4	8.0			33.8	18.9			33.8	24.2			27	164	
	R			215.6	7.4			215.5	−3.4			215.5	−4.5			25	162	
	C			35.1	−3.6			35.2	7.2			35.2	9.4			28	162	10
KJ	N	275	30	20.5	18.6			19.9	−10.3			19.9	−14.2			20	181	13
KK	N	269	5	22.5	12.7			22.1	8.1			22.1	13.1			33	176	
NK	N	324	5	28.3	15.3			28.8	10.8			28.8	16.1			33	167	12
KT	N	271	30	356.1	27.6			356.7	−2.3			356.7	−2.9			27	206	
LJ	N	0	0	36.9	−13.8			36.9	−13.8			36.9	−20.1			13	165	
	R			204.0	−20.2			204.0	−20.2			204.0	−28.7			41	171	
	C			32.5	−2.3			32.5	−2.3			32.5	−3.4			22	166	21
HA, HC, HV	R	150	41	216.8	33.4			220.7	−5.0			220.7	−13.3			28	155	21
	N			24.0	9.7	10	26	23.8	1.7	10	28	23.7	2.6	12	20	28	175	8
Mean	R			212.0	6.9	45	9	213.6	−9.6	19	42	213.7	−15.6	23	31	31	162	17
	C			24.9	5.5	12	17	25.2	2.1	8	37	25.2	3.9	9	29	28	173	7

Pol. is polarity; N, R, C = normal, reversed, combined. Az. and dip are the dip azimuth and dip of the bedding (this work and data by Hannu Kujala, Geological Survey of Finland). *D*, *I* = declination, inclination of the remanent magnetization component; α_{95} (°) = the radius of the 95% confidence circle of the mean direction; *k* = precision parameter (Fisher, 1953); A_{95} = the radius of the of the 95% confidence circle of the paleomagnetic pole.

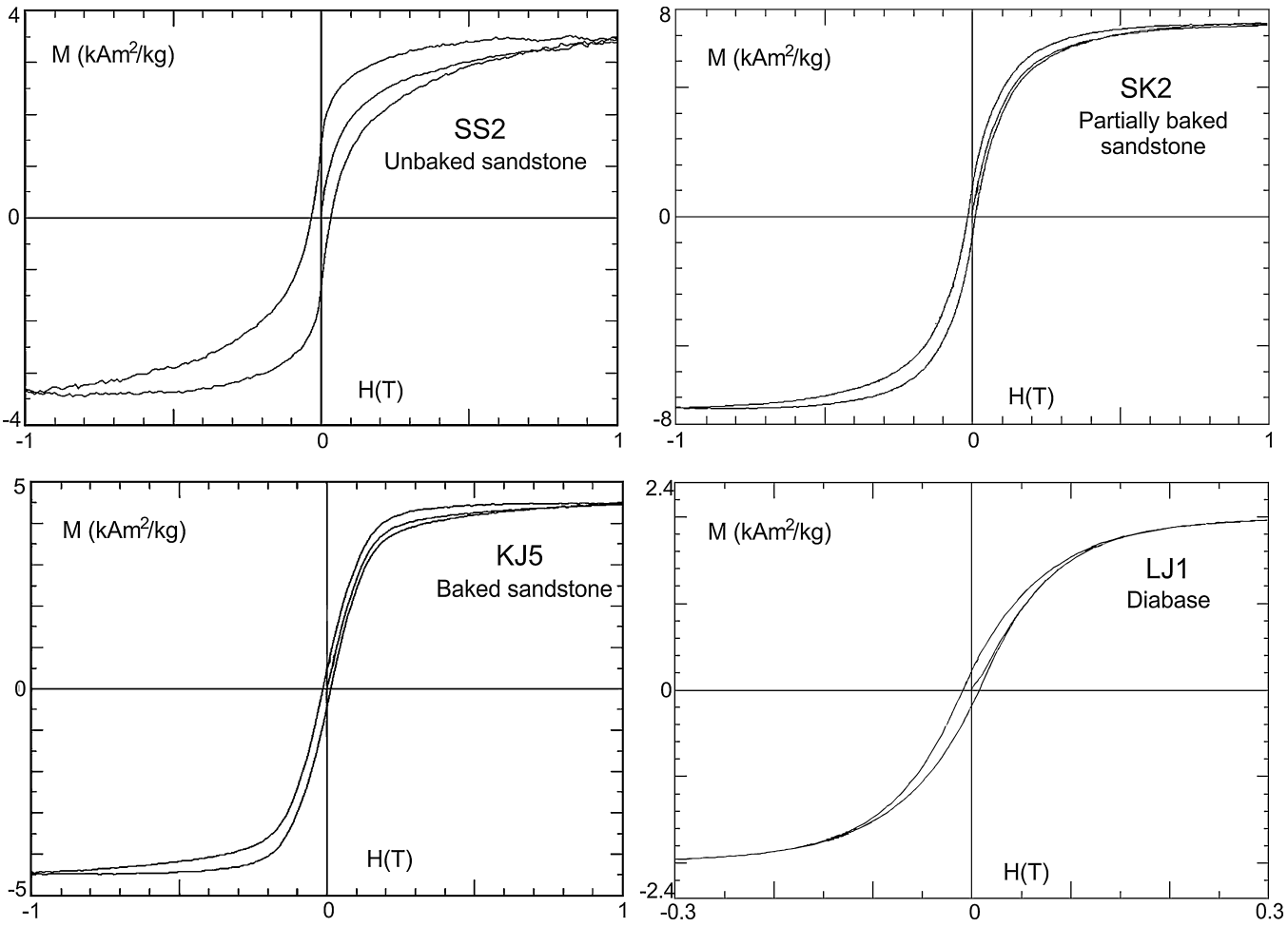


Fig. 4. Hysteresis loops for Satakunta sandstone and diabase samples.

appears slightly shallower than the diabase component (Fig. 8), we have combined them into an overall grand mean ($D = 37.4^\circ$; $I = -45.2^\circ$; $\alpha_{95} = 6.2^\circ$; $k = 27.7$; 21 sites).

When comparing SS pole with Mesoproterozoic poles of Baltica (Fig. 9) it plots between the 1780 Ma and 1452 poles and overlaps with the 1630 Ma mean Subjotnian quartz porphyry dyke pole (Neuvonen, 1986; Mertanen and Pesonen, 1995; Buchan et al., 2000), implying that the magnetization of the Satakunta sandstone could be of similar age. The mean Postjotnian pole (Pesonen et al., 2003) and Satakunta Postjotnian pole (this work) (Table 4) are both tightly constrained with similar Van der Voo quality, yet, there is a noteworthy difference between them. The Postjotnian pole is the mean direction of three Postjotnian formations: Vaasa, Satakunta and Märket dolerites. The Vaasa dolerites (Neuvonen, 1966) show the lowest inclination ($I_R = -23^\circ$). The Satakunta dolerites by Neuvonen (1965) also have a lower inclination ($I_R = -27^\circ$) compared to the inclination of the combined diabase direction (DB + SS_B) of this work ($I_R = -40^\circ$). The Satakunta dolerite site means of Neuvonen (1965) ranges from $I = -19^\circ$ to $I = -47^\circ$ compared to the site means of this work which ranges from $I = -29^\circ$ to $I = -66^\circ$. The sites of Neuvonen (1965) extents further to the South into the rapakivi granite, compared to the diabase site of this study, which is confined to the sandstone area.

The plotted paleomagnetic poles of the Satakunta sandstone sites (Fig. 9) support the concept by Pokki (2007) that the outcrops in the SW are older than the northeastern outcrops, due to an overall tilting of ca. 5° of the Satakunta sandstone formation. The individual site mean paleomagnetic poles show a pattern, going from sites located in the south and southwestern (older) part of the sandstone formation towards sites in the northwestern (younger) part (Fig. 9B). Collecting samples from a thick stratigraphic section gives confidence that paleosecular variation has been averaged out (Kodama, 2012). However, the between-sample dispersion (S) of seven Satakunta sandstone site poles was plotted on the Model G fit Precambrian secular variation curve (Smirnov et al., 2012). The S-values of each of the sites plot on both sides of the Precambrian S-curve at their corresponding near equatorial latitudes (Fig. 10A). This shows that the plotted site poles (Fig. 9B) do not display paleo-secular variation and we therefore interpret Fig. 9B as the approximate stratigraphic order of the sandstone sites. Fig. 10B shows the approximate stratigraphic order of the sandstone sites based on the plotted site poles shown in Fig. 9B. Furthermore, a mineralogical and chemical study done on Satakunta sandstone by Pokki et al. (in press) also indicates that deposits in the SW are older than the deposits in the NE. This could imply that the geomagnetic reversal took place during a later stage of deposition as it is only the younger site poles (poles towards the northeastern edge of the formation, e.g. Harjavalta) that carry the reversed SS component. The S-values of the diabase and baked sandstone data fit reasonably well to the Precambrian S-curve at moderate latitudes (Fig. 10A).

Analyzed detrital zircons of sandstone show exclusive Svecofennian provenance, dominated by 1.88 Ga synorogenic granitoids, and no evidence of 1.59–1.54 Ga Rapakivi granite adjacent to the sandstone formation (Pokki et al., in press). Therefore, the sandstone was most likely deposited before the exhumation of

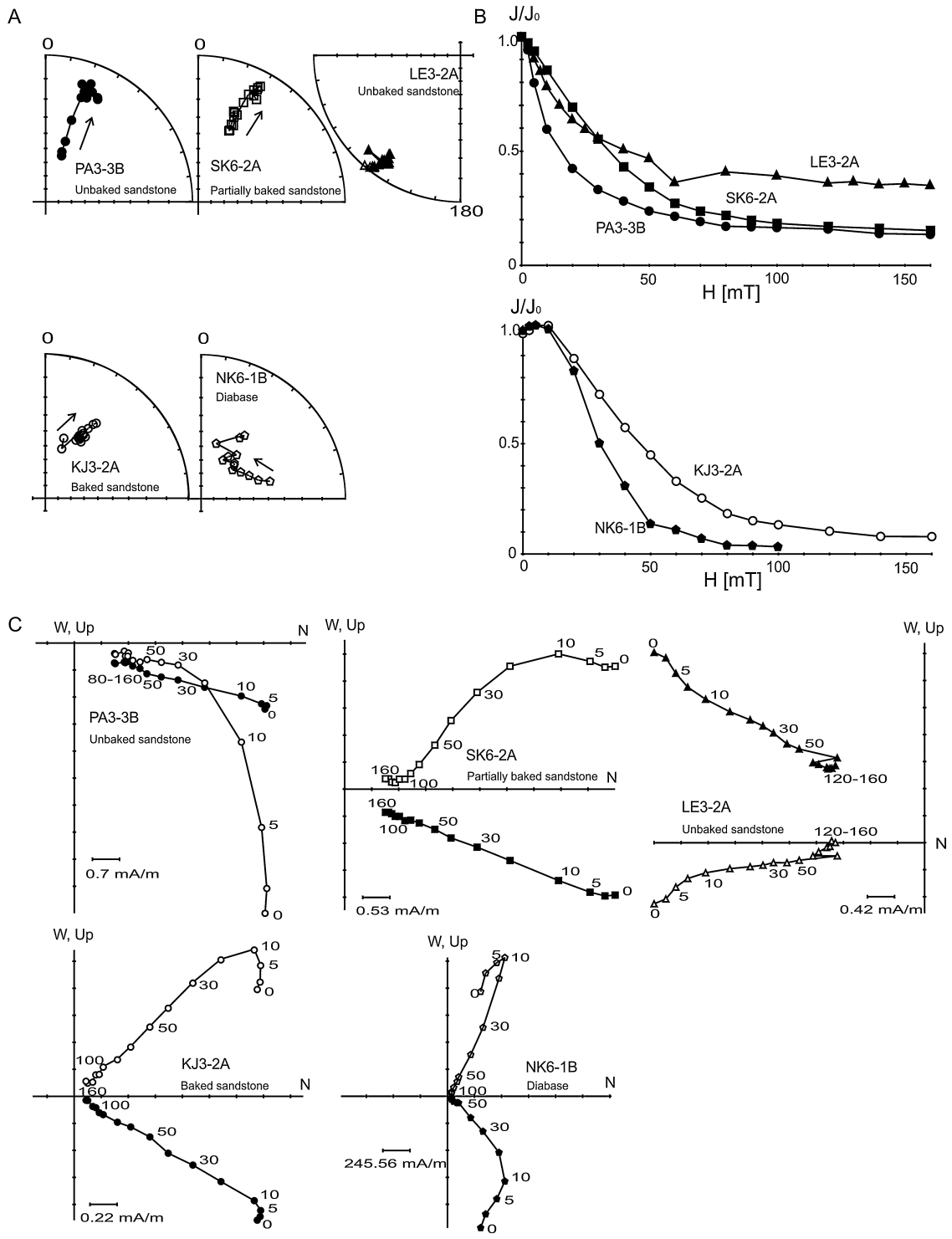


Fig. 5. Examples of AF demagnetization of sandstone specimens. (a) Stereoplots. Arrows show the direction of demagnetization. (b) Relative intensity decay plots. (c) Zijderveld demagnetization plots. Open (closed) symbols indicates vertical (horizontal) plane and numbers indicate demagnetization step in mT. Specimen PA3-3B represents the behaviour of unbaked samples, specimen SK6-2A represents the behaviour of partially baked samples carrying both the primary sandstone and the baked components and specimen KJ3-2A represents completely baked samples. Specimen LE3-2a is an example of a reversed primary component and specimen NK6-1A is an example of a diabase sample.

the Laitila rapakivi batholith and surrounding rapakivi plutons. However, it is not certain if the sandstone deposition is older or younger than the crystallization of the rapakivi batholith. The thermomagnetic curves (Fig. 3) of unbaked and partially baked sandstone show significant changes upon heating, suggesting that the rapakivi event had no, or little effect on the remanent magnetization of the sandstone. This supports the notion that the rapakivi is older than the sandstone, or that the sandstone was at a sufficient distance from the rapakivi not to be remagnetized. Pokki et al. (in press) propose a model in which the basin evolution starts with the rapakivi emplacement, and in which the SW margin is considered the initial deposits during the very early stage of extension. A detailed paleomagnetic study on the Laitila rapakivi batholith could help to clarify the age order of the rapakivi and the sandstone.

Table 4
Selected Proterozoic poles for Baltica.

	Rock unit	Age (Ma)	Pol	D_R	I_R	Plat (°N)	Plon (°E)	A_{95} (°)	Voo 123456 Σ	Ref
1	Mean Svecofennian	1880	N	340	31	41	233	5		1
2	Småland intrusives	1785–1770	C	18	39	46	183	8	111111 6	2
3	Subjotnian quartz porphyry dykes	1630	N	27	12	29	177	6	111010 4	3
4	Satakunta sandstone	ca. 1600	C	30	11	28	173	7	011111 5	This work
			N	29	11	28	178	8		
			R	219	−22	−31	342	17		
5	Kumlinge–Brändö dykes	1576 ± 13	R	206	22	12	182	7	110110 4	4, 5
6	Satakunta sub-Jotnian dykes	1565	C	17	9	29	188	7	111111 6	6
			N	21	16	32	183	7		
			R	184	4	−24	23	11		
7	Föglö–Sottunga dykes	1540 ± 12	N	18	7	28	188	9	110110 4	4, 5
8	Valaam monzodioritic sill	1458	N	40	−11	14	166	2	111010 4	7
9	Lake Ladoga mafic rocks	1452 ± 12	C	30	−15	15	177	6	111111 6	8
10	Mean post-Jotnian intrusions	1265	N	51	−24	4	158	4	111110 5	1
11	Satakunta post-Jotnian (diabase + baked component)		N	43	−40	−3	169	6	111110 5	This work
	Additional poles									
	Shosksha sandstone	1770	C	349	27	40	221	4	111111 6	9
	Dala sediments	1600	C	27	20	33	176	10	001011 3	10
			N	19	15	32	185	21		
			R	213	−23	−33	348	13		
	Åland rapakivi granite	1570	N	13	−11	20	194	16	100000 1	11
	Tuna dykes	1474	N	35	−1	21	171	10	111010 4	12
	Nordingrå gabbro	1552 ± 18	C	56	43	37	138	5	111001 4	13, 14

Age = rock age; Pol = polarity (N, R, C are normal, reversed and combined), D_R and I_R are declination and inclination, respectively, of the mean magnetization calculated for a common reference locality at Kajaani (64.1° N, 27.7° E); Plat/Plon = pole latitude/longitude, A_{95} = confidence circle of the pole, Voo is the Van der Voo reliability criteria of paleomagnetic poles (Van der Voo, 1990). References: (1) Pesonen et al. (2003), (2) Pisarevsky and Bylund (2010), (3) Buchan et al. (2000), (4) Pesonen and Neuvonen (1981), (5) Suominen (1991), (6) Salminen et al. (2014), (7) Salminen and Pesonen (2007), (8) Lubnina et al. (2010), (9) Pisarevsky and Sokolov (2001), (10) Piper and Smith (1980), (11) Johnson (1979), (12) Piper (1992), (13) Piper (1980), and (14) Welin and Lundqvist (1984).

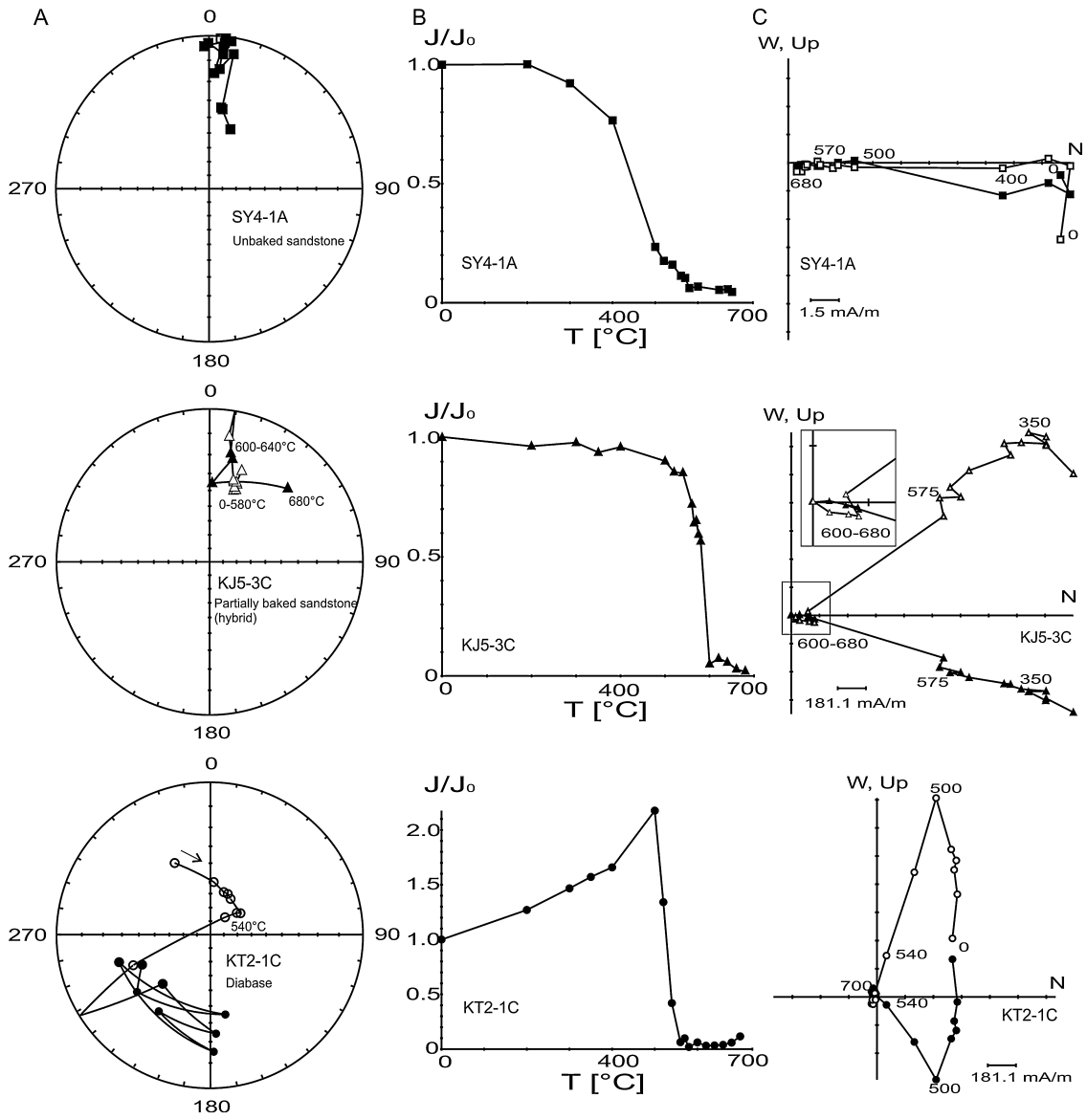


Fig. 6. Examples of thermal demagnetization of an unbaked and partially baked sandstone specimens and a diabase dyke specimen. (a) Stereoplots, (b) relative intensity decay plot and (c) Zijderveld orthogonal plots. Open (closed) symbols indicates vertical (horizontal) plane and numbers indicate demagnetization step in °C.

5.2. Unity of Baltica and Laurentia during the Mesoproterozoic

The unity of Baltica and Laurentia, during 1.8–1.2 Ga, formed the core of the Columbia supercontinent (Rogers and Santosh, 2002; Pesonen et al., 2003, 2012; Zhao et al., 2004; Evans and Mitchell, 2011). Many authors propose the NENA (North Europe–North America) configuration, in which northern Greenland is positioned alongside northern Norway (e.g. Gower et al., 1990; Salminen and Pesonen, 2007; Evans and Pisarevsky, 2008; Evans and Mitchell, 2011; Salminen et al., 2014). This is supported by a tight cratonic fit, concordance with basement geology, correlations of geochronology, and matching of recent paleomagnetic poles. If paleomagnetic poles from two or more adjacent continents overlap, it verifies the validity of the configuration for the proposed time interval; therefore the NENA configuration is valid at 1.75 Ga, 1.46 Ga and 1.27 Ga. Salminen et al. (2014) also establishes its validity at 1.63 Ga and at 1.56 Ga. Furthermore Archaean to Paleoproterozoic formations are continuous from North America trough Greenland to northern Baltica (Fig. 11). 1.7 Ga subduction-related magmatic formations and 1.27 Ga dykes become contiguous in these reconstructions (Pesonen et al., 2003), also supported by magnetic anisotropy results by Elming and Mattsson (2001).

We rotated Laurentia to the present reference frame of Baltica in the NENA configuration using an Euler pole of Elat = 47°, Elong = 1.5°, angle = −49° (Evans and Pisarevsky, 2008) and rotated this assembly using the paleomagnetic pole of Satakunta sandstone, to the actual paleolatitude. The Euler poles resulting from this rotation was used to rotate selected paleomagnetic poles of Baltica and Laurentia in order to compare the respective APWPs (Table 5). In this reconstruction the combined mean pole of Satakunta Sandstone (B1600) does not overlap with the 1592 ± 3 Ma Western Channel Diabase (L1590) pole. This can be due to a possible age difference between the two formations. However, as with the APWP, the Satakunta sandstone pole shows a significant overlap with the 1630 Ma mean Subjotnian quartz porphyry dyke pole (B1630) (Neuvonen, 1986; Mertanen and Pesonen, 1995; Buchan et al., 2000).

The configuration of NENA shown in Fig. 11 looks similar to the 1.63 Ga reconstruction of Baltica and Laurentia of Pesonen et al. (2012). In that reconstruction the Melville Bugt (1630 Ma) pole (L 1630) was used to position Laurentia by applying the

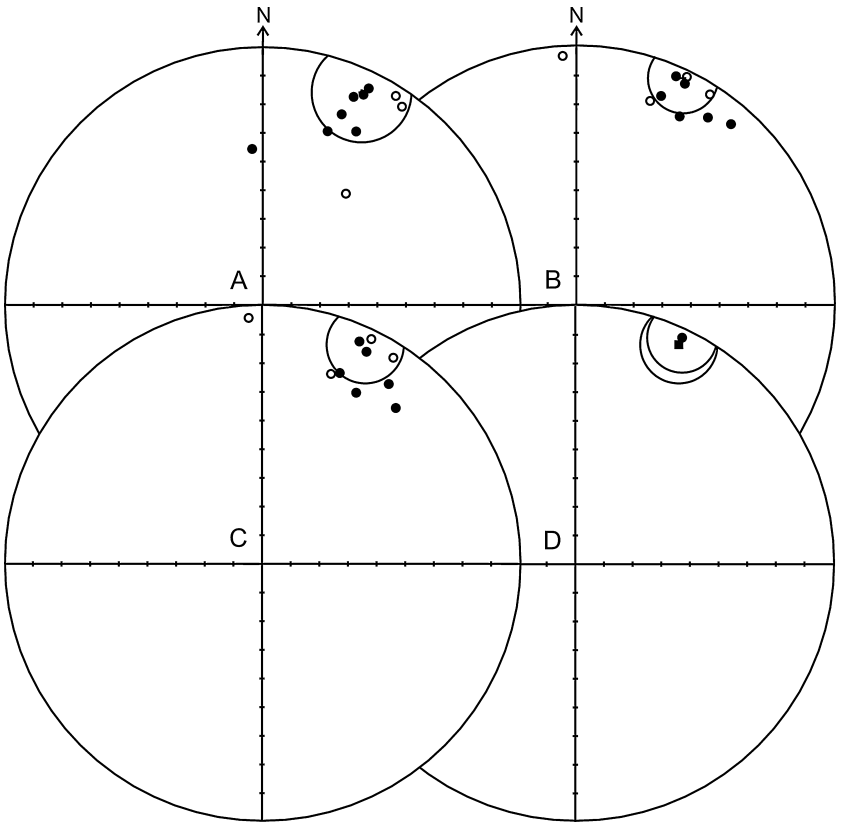


Fig. 7. Site mean directions of the primary sandstone component (A) before tilt correction (D_m = 24.9°; I_m = 5.5°; α_{95} = 12.0; k = 17.2), (B) after tilt correction (D_m = 25.2°; I_m = 2.1°; α_{95} = 8.0; k = 37.2) and (C) after inclination shallowing correction (D_m = 25.2°; I_m = 3.9°; α_{95} = 9.1°; k = 28.9). (D) Mean direction before (circle) and after (square) inclination correction.

closest approach technique. In the APWP matching of Laurentia and Baltica (Fig. 11), the Melville Bugt pole (L1630) plots far from the Satakunta sandstone pole as well as the Western Channel dykes pole (L1590). As discussed by Salminen et al. (2014), this can be due to a large scatter of the Melville Bugt site poles or it could be related to the uncertainty of the age of Melville Bugt dykes, since no baked contact test was performed.

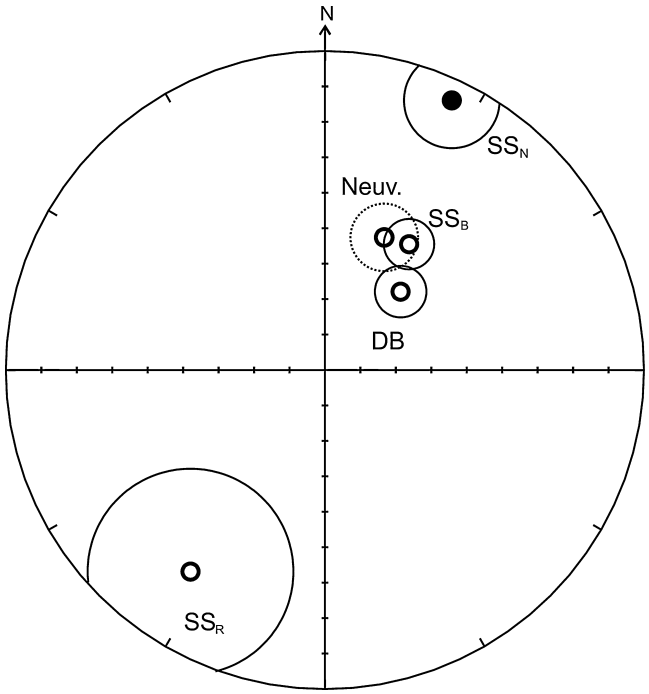


Fig. 8. Mean paleomagnetic directions for sandstone normal polarity (SS_N), sandstone reversed polarity (SS_R), sandstone baked component (SS_B) and diabase (DB) component. Neuv. represents the paleomagnetic direction for sandstone by Neuvonen (1973).

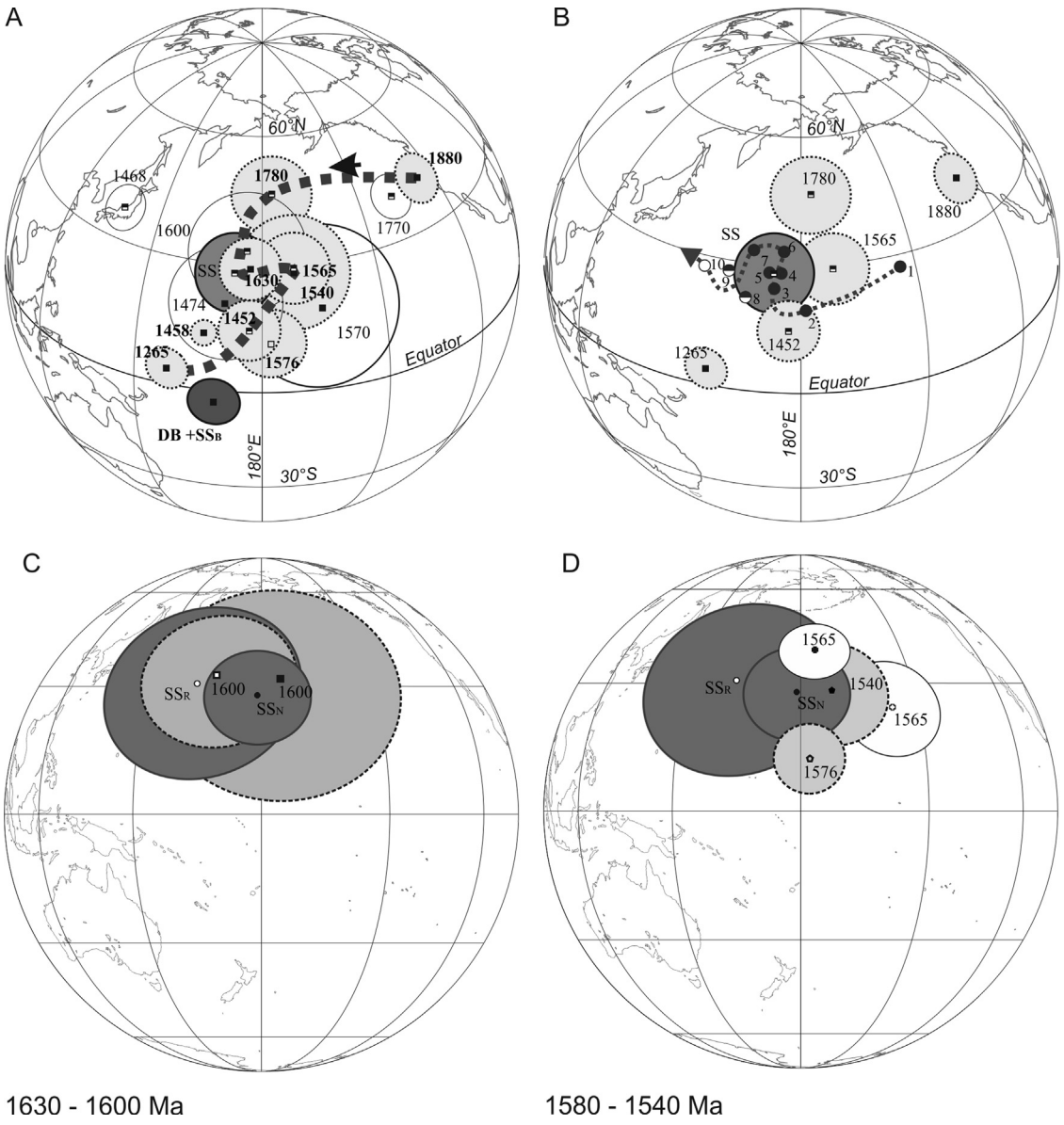


Fig. 9. (A) Paleomagnetic poles of this study with 95% confidence circle (darker grey) plotted with key and other significant paleomagnetic poles (lighter grey), summarized in Table 4. Closed (open) squares represent normal (reversed) polarity, and mixed squares represents mixed polarity. Poles SS, SS_B and DB are the primary sandstone pole, the baked sandstone pole and the Satakunta diabase pole, respectively. Poles with open confidence circles are additional pole from Table 4. (B) Paleomagnetic poles of the Satakunta sandstone sites, plotted with selected paleomagnetic key poles. Sites are plotted in the following order: 1. KT, 2. KJ, 3. SK, 4. MU, SS, SY, 5. PA, 6. KK, 7. NK, 8. LJ, 9. LE, LS, 10. HA, HC, HV. (C) Normal and reversed poles of the Satakunta sandstone pole plotted with normal and reversed poles of Dala sandstone (ca. 1600 Ma), and (D) with normal and reversed poles of Satakunta subjotnian dykes (1565 Ma), Kumlinge–Brändö dykes (*R*) (1576 ± 13 Ma) and Föglö–Sottunga dykes (N) (1540 ± 12 Ma).

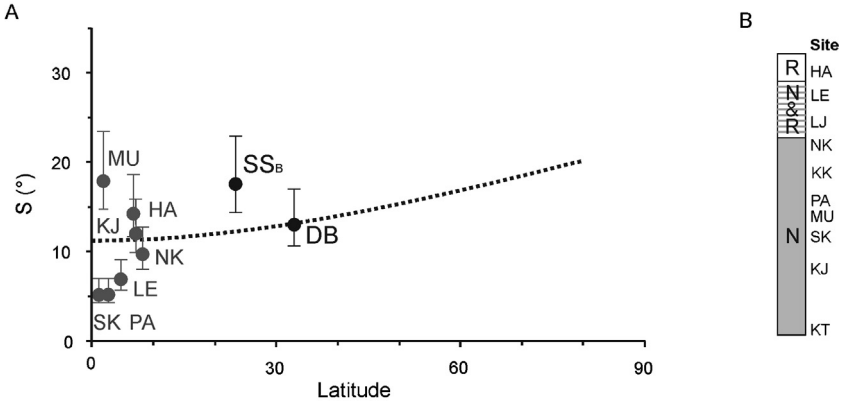


Fig. 10. (A) Average of between sample dispersion of Satakunta sandstone poles and between site dispersion of baked sandstone and diabase poles plotted on a Mesoproterozoic secular variation model by Smirnov et al. (2012). (B) Approximate stratigraphy of the Satakunta sandstone sites based on APWP of sites shown in Fig. 9B. N and R are normal and reversed polarity, respectively.

Table 5
Paleomagnetic poles used for reconstruction.

Craton	Formation	Plat (°)	Plon (°)	Rplat (°)	Rplon (°)	A_{95}	Age (Ma)	References
B	Småland intrusives	46	183	70	332	8	1785–1770	1
L	Cleaver Dykes	19	277	40	308	6	1740 + 5/−4	2
B	Satakunta sandstone	28	173	90	0	7	ca. 1600	This work
B	Sub-Jotnian quartz porphyry dykes	29	177	86	280	6	1630	3
L	Melville Bugt dyke swarm[a]	2	262	50	279	9	1630	4
B	Satakunta Subjotnian	29	188	77	271	7	1565	5
L	Western Channel Diabase	9	245	68	278	7	1590 ± 4; 1592 ± 2	6, 7
B	Lake Ladoga mafic rocks	15	177	76	190	6	1452 ± 12	8
L	Mean	−7	218	65	193	6	1468	5
B	Mean post-Jotnian dolerite intrusions	4	158	62	140	4	1265	9
L	Mackenzie dykes	4	190	54	136	5	1267 ± 2	10

B is Baltica, L is Laurentia; Plat/Plon is the pole latitude/longitude and A_{95} is the confidence circle of the pole.

[a] Greenland pole has been rotated to reference frame of Laurentia using Euler parameters: ELat: 67.5, Elon: 241.5 and Erot: −13.8 (Roest and Srivastava, 1989). Rplat/Rplon is the pole latitude/longitude presented in Fig. 11. References: (1) Pisarevsky and Bylund (2010), (2) Irving et al. (2004), (3) Buchan et al. (2000), (4) Halls et al. (2011), (5) Salminen et al. (2014), (6) Irving et al. (1972a), (7) Hamilton and Buchan (2010), (8) Lubnina et al. (2010), (9) Pesonen et al. (2003), and (10) Buchan and Halls (1990).

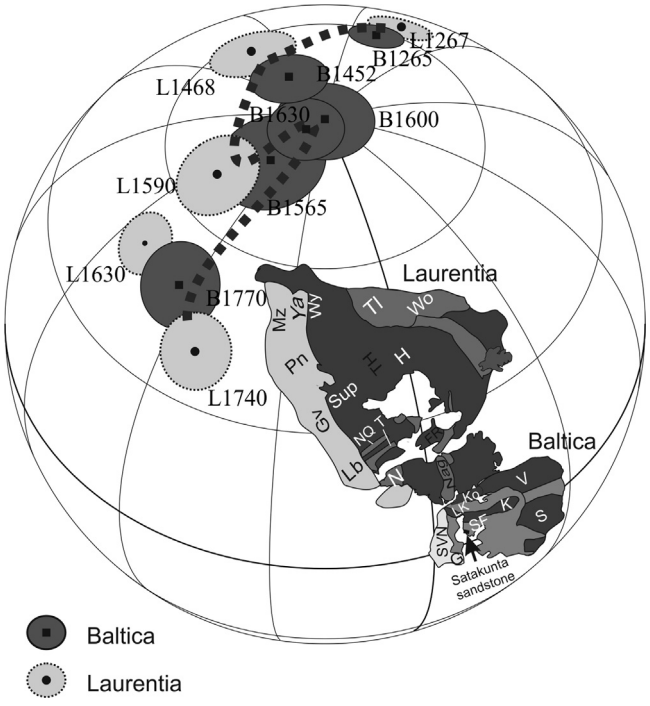


Fig. 11. Reconstruction of Baltica and Laurentia at 1.6 Ga based on mean poles for Baltica and Laurentia (Table 5). Euler pole for Baltica is Lat: 0°, Lon: 263°, angle −62°, and for Laurentia Lat: 11.3°, Lon: 305.8°, angel: −73.7°. Poles for Baltica (B) and Laurentia (L) shown with age (Ma).

6. Conclusions

Paleomagnetic measurements on Satakunta sandstone revealed a ChRM of $D = 25.2°$; $I = 3.9°$ ($\alpha_{95} = 9.1$, 10 sites). The primary origin of this direction is supported by a positive reversal test and a positive tilt correction and an inverse baked contact test. The paleomagnetic pole of this direction (26.8° N, 173.5° E; $A_{95} = 5.7°$) is close to the 1.63 Ga from Sipoo dykes (Mertanen and Pesonen, 1995), with the A_{95} circles overlapping significantly. We therefore estimate the age of the Satakunta sandstone at ca. 1.6 Ga. However, because the pole also overlaps with poles with the ages of 1452–1630 Ma, the age of the sandstone cannot be accurately defined. The reconstruction of Baltica and Laurentia by using the Satakunta sandstone pole confirms the NENA configuration.

A baked paleomagnetic component was isolated from 13 sites ($D = 34°$; $I = -40.7°$; $\alpha_{95} = 8.1°$). This component is very close to the one of nearby diabase sites, and we therefore consider this component to be caused by baking. Therefore the component reported as the Satakunta sandstone component by Neuvonen (1973), is most likely a baked component.

Acknowledgments

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